

A Complete Course in Chemistry

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7 December, 2018

Physical Chemistry

Part I

Stoichiometry (Quantitative Analysis)

Chapter 1

Governing Laws

1.1 The law of definite proportions

The law of definite proportions, also known law of definite composition, states that regardless of the amount, a pure compound always contains the same elements in the same proportions by mass.

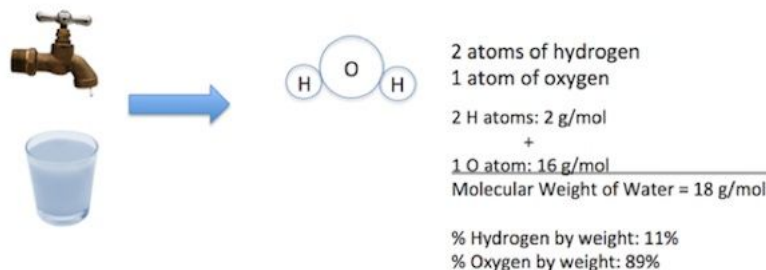
1.1.1 History

Although many experimenters had long assumed the truth of the principle in general, the French chemist Joseph-Louis Proust first accumulated conclusive evidence for it in a series of researches on the composition of many substances, especially the oxides of iron (1797). Another French chemist, Claude Berthollet, who held for indefinite proportions, contested Proust's findings, but the Scottish chemist Thomas Thomson confirmed some of them. Proust had definitely proved "metals are not capable of indefinite degrees of oxidation." The principle was then concretely formulated by the English chemist John Dalton in his chemical atomic theory (1808).

1.1.2 Examples of the Law of Definite Proportions

Let us take, for example, the compound water. Whatever the source of water, its composition is that of two atoms of hydrogen and one atom of oxygen.

Law of Definite Proportions: % by Weight of Hydrogen and Oxygen in Water This figure shows that water, from any source, is always made up of two atoms of hydrogen and one atom of oxygen. If we calculate the molecular weight of water, we come up with 18 g/mol. In 1 mole of water, there are 2 grams of hydrogen and 16 grams of oxygen. By weight, we have a percentage of 11% hydrogen and 89% oxygen in one mole of water. This translates into a ratio of 1:8 of hydrogen to oxygen in water.

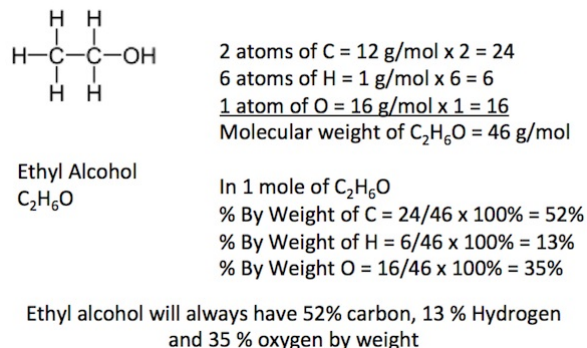


Law of Definite Proportions: % by Weight of Hydrogen and Oxygen in Water

The Law of Definite Proportions illustrates that whatever the amount of water, whether it be 2 moles or 54 grams, the ratio of the amount of hydrogen to oxygen by weight will always be the same, just like the egg:sugar:butter ratio in chocolate cake should always be the same to maintain consistent taste.

Let's look at another compound made up of three atoms this time, such as Ethyl Alcohol, which consists of 2 atoms of Carbon, 6 atoms of Hydrogen and one atom of Oxygen.

According to the figure, we can calculate the percentage of each element in ethyl alcohol by weight, and we come up with 52% carbon, 13% hydrogen and 35% oxygen by weight. Based on the Law of Definite Proportions, the percentages of carbon, hydrogen and oxygen in ethyl alcohol, will always be the same, no matter what the amount.



1.2 Law of multiple proportions

Law of multiple proportions, also known as Dalton's Law, states that when one element combines with another to form more than one compound, the mass ratios of the elements in the compounds are simple whole numbers of each other.

Part II

States of Matter

Chapter 2

The Solid State

Test Solid State (NCERT)

Time Allowed : 1 Hour _____ **Maximum Marks : 60**

Please read the instructions carefully. You will be allotted 5 minutes specifically for this purpose.

Instructions

A. General

1. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers, and electronic gadgets in any form are not allowed.
2. Do not break the seals of the question-paper booklet before instructed to do so by the invigilators.

B. Question paper format and Marking Scheme :

1. This question paper consists of 6 questions carrying 10 marks each.

Q1: Classify each of the following solids as ionic, metallic, molecular, network (covalent) or amorphous.

- (i) Tetra phosphorus decoxide (P_4O_{10})
- (ii) Ammonium phosphate ($(NH_4)_3PO_4$)
- (iii) SiC
- (iv) I_2
- (v) P_4
- (vi) Plastic
- (vii) Graphite
- (viii) Brass
- (ix) Rb
- (x) LiBr
- (xi) Si

Q2: Silver crystallises in fcc lattice. If edge length of the cell is 4.07×10^{-8} cm and density is 10.5 g cm^{-3} , calculate the atomic mass of silver.

Q3: Analysis shows that nickel oxide has the formula $Ni_{0.98}O_{1.00}$. What fractions of nickel exist as Ni_{2+} and Ni_{3+} ions?

Q4: Ferric oxide crystallises in a hexagonal close-packed array of oxide ions with two out of every three octahedral holes occupied by ferric ions. Derive the formula of the ferric oxide.

Q5: Explain the following terms with suitable examples:

- (i) Schottky defect
- (ii) Frenkel defect
- (iii) Interstitials and
- (iv) F-centres.

Q6: If NaCl is doped with 10^{-3} mol % of $SrCl_2$, what is the concentration of cation vacancies?

Chapter 3

The Gaseous State

Part III

Redox Reactions and Electrochemistry

Part IV

Chemical Kinetics and Equilibrium

Part V

Structure of the Atom

Inorganic Chemistry

Part VI

The Salt Analysis (Qualitative Analysis)

Part VII

Common Industrial Extraction and Purification Techniques

Organic Chemistry

Part VIII

Introduction

Chapter 4

Some Facts

Part IX

IUPAC Nomenclature

Part X

Hydrocarbons

Part XI

Stereochemistry

